

Predicting Movie Ratings Based on Metadata

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Abstract—Accurate prediction of movie ratings from publicly available metadata can significantly aid decision-making in the film industry. In this work, we compile a dataset of 7,036 films (1975–2024) collected from IMDb, Kaggle, and GitHub. Features are 23 attributes. Such as, Runtime, release year, budget adjust by inflation, popularity, facebook actor director likely, genre, and etc. After rigorous cleaning, we are using imputation, removal outlier, normalization, and encoding technique to transform data.

We tested multiple MLP architectures (from shallow to deep) and found the best single hidden layer model—a 64-neuron layer with *tanh* activation, trained for 100 epochs with batch size 64—which achieves $MSE = 0.704$ and $R^2 = 0.384$ on a held-out test set. We also compared against a Random Forest baseline, which yields $MSE = 0.6186$ and $R^2 = 0.4589$ on its test split, with 5-fold cross-validation reporting $MSE = 0.6832 \pm 0.0476$ and $R^2 = 0.4256 \pm 0.0253$.

While the ANN provides valuable insights into hyperparameter impacts. However the random tree-based methods demonstrate superior predictive power on tabular movie metadata. Our findings offer practical guidelines for selecting and tuning regression models in entertainment analytics, balancing architectural complexity against generalization ability.

Keywords: Movie Rating Prediction, Deep Learning, Regression, MLP-ANN, Random Forest, IMDb, Kaggle, Github

I. INTRODUCTION: TODO - ADD INFORS OF FACEBOOK METADATA

The prediction of movie ratings has become an increasingly important task in the era of digital entertainment, where user-generated reviews and ratings play a critical role in shaping audience preferences, marketing strategies, and content production decisions. Accurate rating prediction models can benefit a wide range of stakeholders, including film studios, streaming platforms, and recommendation systems, by enabling more informed decisions regarding movie production, acquisition, and promotion.

Movie ratings, such as those provided by IMDb, are influenced by a multitude of factors, including cast, crew, budget, genre, language, and release timing. The complexity of these relationships, combined with the heterogeneity and incompleteness of real-world data, presents significant challenges for predictive modeling. Traditional approaches, such as collaborative filtering and content-based methods, often fall short in capturing the nuanced interactions among diverse metadata features, especially for movies that have not yet been released or lack sufficient historical data.

Recent advances in machine learning, particularly in feature engineering and deep learning, offer new opportunities to address these challenges. By integrating and preprocessing

data from multiple sources—including IMDb, Kaggle, and external budget datasets—and engineering features that reflect the impact of actors, directors, genres, and financial investment, it is possible to build more robust and accurate predictive models. In this project, we develop a comprehensive machine learning pipeline that leverages both traditional regression techniques and artificial neural networks to predict IMDb movie ratings prior to release. Our approach emphasizes rigorous data cleaning, inflation adjustment of financial features, and the use of advanced feature engineering to maximize predictive performance and interpretability.

The remainder of this paper is organized as follows: Section II describes the problem statements and explain why this research is important; Section III presents exploratory data analysis and feature selection; Section IV details the modeling approach and evaluation metrics; Section V discusses experimental results and comparative analysis; and Section VI concludes with insights and directions for future work.

II. DATASET DESCRIPTION

1) *Data Sources*: We compiled our dataset by merging three publicly available sources: the IMDb Non-Commercial Datasets[1], Kaggle movie metadata[2], and social media metadata from GitHub[3]. IMDb provides core movie features including ratings, titles, and cast information. Kaggle contributes financial and popularity data such as budgets and revenues. The Facebook metadata offers user engagement metrics, including director and actor Facebook likes.

2) *Merging Strategy*: The IMDb and Kaggle datasets were merged using the shared `imdb_id` field. For the Facebook dataset, we extracted this identifier from the `movie_imdb_link` field, which stores the IMDb URL (e.g., `http://www.imdb.com/title/tt0499549/...`). The identifier substring (e.g., `tt0499549`) was parsed and matched to the existing `imdb_id` field.

3) *Budget Inflation Adjustment*: To ensure consistency in financial comparisons across decades, all movie budgets were adjusted for inflation using the Consumer Price Index (CPI). We used CPI data from the U.S. Bureau of Labor Statistics[4], converting all values to their equivalent in 2025 dollars. This adjustment reduces temporal bias and improves the interpretability of financial features.

4) *Final Dataset Summary*: After merging and preprocessing—including handling missing values, removing outliers, and encoding categorical variables—the final dataset contains over 7,000 movies, as shown in Figure 1. Key features

include genres, cast, crew, inflation-adjusted budgets, runtime, language, release year, and social media indicators.

```
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RangeIndex: 9335 entries, 0 to 9334
Data columns (total 41 columns):
#   Column                Non-Null Count  Dtype
---  -
0   id                     7741 non-null   float64
1   belongs_to_collection  1575 non-null   object
2   budget                 8991 non-null   float64
3   genres                 7741 non-null   object
4   homepage               2532 non-null   object
5   imdb_id                9335 non-null   object
6   original_language      7741 non-null   object
7   original_title         7741 non-null   object
8   overview               7719 non-null   object
9   popularity             7741 non-null   float64
10  poster_path            7739 non-null   object
11  production_companies   7741 non-null   object
12  production_countries   7741 non-null   object
13  release_date           7740 non-null   object
14  runtime                7741 non-null   float64
15  spoken_languages       7741 non-null   object
16  status                 7741 non-null   object
17  tagline                6275 non-null   object
18  title                  9332 non-null   object
19  Keywords               7067 non-null   object
20  cast                   7741 non-null   object
21  crew                   7741 non-null   object
22  revenue                6090 non-null   float64
23  tconst                 7739 non-null   object
24  averageRating          9333 non-null   float64
25  release_year           9223 non-null   float64
26  release_month          7740 non-null   float64
27  release_day            7740 non-null   float64
28  budget_adjusted        7741 non-null   float64
29  budget_adjusted_log    7741 non-null   float64
30  director_name          5197 non-null   object
31  num_critic_for_reviews 5253 non-null   float64
32  director_facebook_likes 5197 non-null   float64
33  actor_1_name           5298 non-null   object
34  actor_1_facebook_likes 5298 non-null   float64
35  actor_2_name           5292 non-null   object
36  actor_2_facebook_likes 5292 non-null   float64
37  actor_3_name           5282 non-null   object
38  actor_3_facebook_likes 5282 non-null   float64
39  num_user_for_reviews   5284 non-null   float64
40  movie_facebook_likes   5305 non-null   float64
dtypes: float64(18), object(23)
memory usage: 2.9+ MB
```

Fig. 1. Distribution of IMDb Ratings. The dataset consists of 9,335 movies and 41 features sourced from IMDb, Kaggle, and Facebook metadata.

III. EXPLORATORY DATA ANALYSIS (EDA)

This section presents a systematic exploration of the dataset to uncover structural patterns, identify data quality issues, and guide preprocessing decisions. The dataset comprises 7,036 movies and includes 23 metadata features spanning numeric, categorical, and temporal types. Also we are following with Purohit’s EDA [5, EDA Cleaning Process] concept to clean-up datasets.

1) *Null and Zero-Value Analysis:* We began the exploratory analysis by quantifying data completeness across all features. For each column, we computed the number of non-null values and the percentage of missing entries. Features with more than 20% missing data will caused potential imputation, transformation, or removable in to the data preprocessing.

The high proportion of zeros in `movie_facebook_likes` (24.5%) suggests that we need cleaning this part of data or

TABLE I
ZERO-VALUE SUMMARY FOR SOCIAL ENGAGEMENT FEATURES

Feature	Zero Count	Zero %
<code>movie_facebook_likes</code>	2285	24.48%
<code>director_facebook_likes</code>	967	10.36%
<code>actor_3_facebook_likes</code>	89	0.95%
<code>actor_2_facebook_likes</code>	55	0.59%
<code>actor_1_facebook_likes</code>	26	0.28%

this data is make sence. Thus, I was remove (50%) data as missing and saving other 50% as some user give movie as really bad score. Based on the real word cased. And other actor missing percentage is not too higher. Thus, we are removed those data.

2) *Avoid Pitfall Issues:* After we are carefully reviewed the table (Table II) feature availability in real-world case. Specifically, we excluded the `revenue` feature from our datasets. Since our goal is to predict IMDb ratings before the movie’s release, but using `revenue`—which is only known as after movie relase. Thus, we are remove it.

TABLE II
SUMMARY STATISTICS (COUNT, STD, MIN, MAX)

Feature	Count	Std	Min	Max
<code>id</code>	7741	2140.68	1.00	7398.00
<code>budget_adj</code>	7741	50087280	1.00	308893700
<code>popularity</code>	7741	6.28	0.000001	45.18
<code>runtime</code>	7741	19.66	41.60	174.12
<code>revenue</code>	6090	1.18e+08	1.00	1.52e+09
<code>averageRating</code>	9333	1.05	1.20	9.50
<code>release_year</code>	9223	16.40	1916	2074
<code>release_month</code>	7740	3.39	1.00	12.00
<code>budget_adj_log</code>	7741	1.80	0.69	19.55
<code>director_fb_likes</code>	5197	2774.46	0	23000
<code>actor_1_fb_likes</code>	5298	14853.72	0	640000
<code>movie_fb_likes</code>	5305	19518.40	0	349000

3) *Outlier Analysis:* We are trying to improve the robustness of modeling and reduce the influence of extreme values, we conducted an outlier analysis on key numeric features using the Interquartile Range (IQR) method. Outliers were defined as values falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.

Table III presents the number of detected outliers per feature and the corresponding IQR-based thresholds. Features such as `budget_adjusted_log`, `director_facebook_likes`, and `actor_2_facebook_likes` showed substantial outlier counts, with extreme values reaching magnitudes significantly larger than the typical data range. These observations were either removed or clipped based on distributional impact and downstream model sensitivity.

Figure 2 This boxplot was visually that which feature’s outlier should be removed. This process will leading us datasets more cleaner and improved the stability, before we are training the models.

4) *Correlation Analysis:* We are used the Pearson correlation matrix to check coorelation with all features. As a results, visualized in Figure 3, reveal several moderate correlations relevant to feature engineering.

TABLE III
OUTLIER SUMMARY BASED ON IQR THRESHOLDS

Feature	Lower Bound	Upper Bound	Outliers
popularity	0.00	21.59	218
runtime	58.00	154.00	286
budget_adj_log	14.70	19.71	591
num_critic_for_reviews	0.00	427.00	198
num_user_for_reviews	0.00	745.50	415
director_fb_likes	0.00	512.00	584
log_actor_1_fb_likes	2.17	13.59	28
actor_2_fb_likes	0.00	1877.50	694
actor_3_fb_likes	0.0	1394.62	200
log_movie_fb_likes	0.42	15.43	0

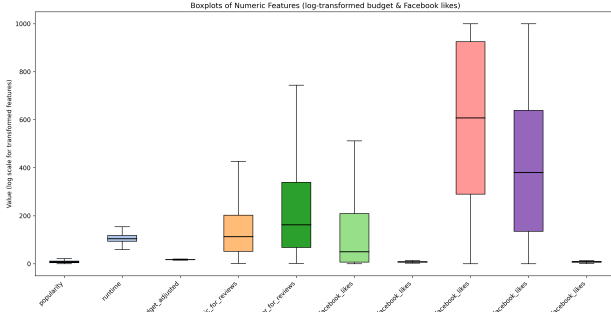


Fig. 2. Boxplot of selected numeric features before outlier removal.

Among the most significant findings:

- `num_critic_for_reviews` and `movie_facebook_likes` exhibit a strong correlation ($r = 0.64$), suggesting shared predictive power or redundancy.
- `num_user_for_reviews` is also strongly correlated with both `num_critic_for_reviews` ($r = 0.58$) and `movie_facebook_likes` ($r = 0.39$).
- `runtime` has moderate correlation with `popularity` ($r = 0.50$), and both show some correlation with `release month`.
- The target variable, `averageRating`, has weak but positive correlations with `runtime` ($r = 0.11$), `num_critic_for_reviews` ($r = 0.29$), and `movie_facebook_likes` ($r = 0.23$).
- `budget_adjusted` is only mildly correlated with `rating` ($r = 0.04$), but more strongly correlated with `num_critic_for_reviews` ($r = 0.35$) and `num_user_for_reviews` ($r = 0.41$).

These findings suggest that there is no single feature that can stronger predicts `averageRating`, but we will drop the correlation values less then 0.

5) *High-Budget Film Analysis:* We are show that the top-ten films with the highest production budgets in the dataset. All listed titles exceed 250 million. Such as, *Speed 2 Cruise Control*, *Waterworld*, and *Titanic*, each budgeted above 300 million. The remaining films—including *The 13th Warrior*, *Wild Wild West*, and *Batman Robin**—fall between 250 million and 275 million.

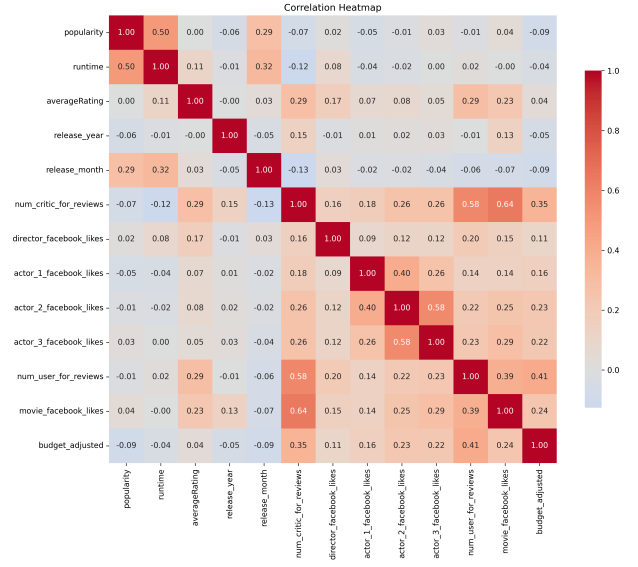


Fig. 3. Pearson correlation heatmap of numeric features.

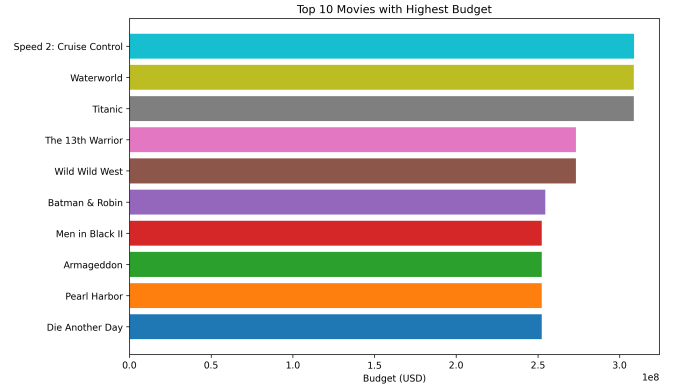


Fig. 4. Top 10 movies by production budget.

Analyzing this upper tail of the budget distribution supports several aspects of exploratory data analysis:

- **Outlier Detection:** These extreme values highlight the need for transformation. To avoid distortion in regression models, a log transformation is applied to the `budget_adjusted` variable, as discussed in Section 3.1.
- **Temporal Context:** Most of these high-budget films were produced in the late 1990s and early 2000s. This trend informs our decision to adjust all budget values for inflation and to account for release-year effects in modeling.
- **Feature Engineering:** The presence of blockbuster films in the dataset motivates the inclusion of proxy variables (e.g., marketing intensity or genre–budget interactions) to better capture variance related to commercial scale.
- **Model Robustness:** Reviewing these outliers informs our decision whether to cap extreme values (e.g., using percentile thresholds) or apply robust modeling techniques

to prevent high-leverage influence from a small number of cases.

This analysis ensures that budget-related features are properly scaled and interpreted, improving generalization and interpretability in downstream predictive models.

- **Outlier Identification:** These extreme budget values constitute the right-hand tail and would disproportionately influence any model that uses raw budgets. We therefore apply a log-transform (see Section 3.1) to stabilize variance and reduce leverage of these outliers.
- **Temporal Context:** All top-budget films are clustered in the late 1990s and early 2000s, revealing a period of escalating production costs. This informs our decision to adjust budgets for inflation (Section 2.4) and to include release year interactions in downstream models.
- **Feature Engineering:** Knowing which films occupy the upper tail allows us to consider additional indicators of blockbuster status—such as marketing spend proxies or genre–budget interactions—in our feature set.
- **Model Robustness:** By inspecting the most extreme cases, we can decide whether to cap budgets at a percentile threshold or to use robust regression techniques that mitigate the influence of these few high-budget titles.

Overall, this analysis of top-budget films informs our transformations, prevents budget-driven skew in the model, and helps us engineer features that capture both absolute and relative production scale.

6) *Genre Distribution and Mean Rating:* Figure 5 shows, for each first-listed genre, the total number of films (blue bars, log-scaled) and the mean IMDb rating (red bars). Drama is the most frequent genre (nearly 900 titles), followed by Comedy and Action (each with approximately 700–800 films). In contrast, genres such as Foreign and History appear in fewer than 20 films.

Despite these large differences in sample size, mean ratings vary only modestly—ranging from about 6.2 to 7.5. Niche categories (e.g., Documentary, Mystery, Foreign) tend to exceed the overall average rating (≈ 6.42), whereas high-volume genres (Drama, Comedy) cluster near the dataset mean. This pattern indicates that no single genre overwhelmingly drives rating outcomes, underscoring the need for multivariate models to capture these nuanced effects.

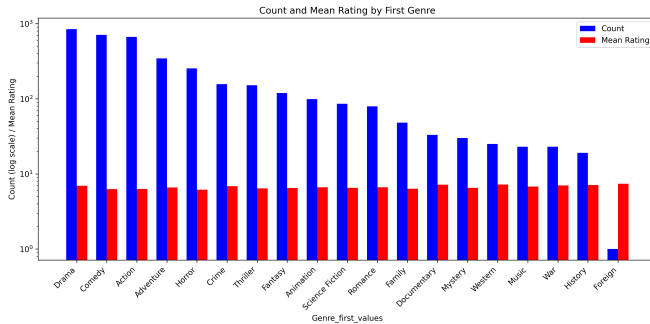


Fig. 5. Count (log scale) and mean IMDb rating by first-listed genre.

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